

Mock Test 1

Read the following with your child:

1. This is a multiple-choice paper, in which you have to mark your answer to each question on the separate answer sheet. You should mark only one answer for each question.
 2. Draw a firm line clearly through the rectangle next to your answer like this . If you make a mistake, rub it out as completely as you can and put in your new answer.
 3. Be sure to keep your place in the correct section of the answer sheet. Mark your answer in the box that has the same number as the question in the booklet.
 4. You may find some of the questions difficult. If you cannot do a question, **do not waste time on it but go on to the next**. If you are not sure of an answer, choose the one you think is best.
 5. **Work as quickly and as carefully as you can.**
-

Familiarisation

1

What is this number in figures:

six hundred and two thousand, one hundred and five

A 62105

B 602150

C 602105

D 620105

E 6002105

2

■ stands for 16 boats.

Look at this table:

Marina A: 3 anchors Marina B: 2 anchors Marina C: $3\frac{1}{2}$ anchors

How many more boats are in Marina C than Marina B?

A 8

B 16

C 24

D 32

E 40

3

What is the value of the 6 in this number?

4.562

A 6 tenths

B 6 hundredths

C 6 thousandths

D 6 tens

E 6 hundreds

4

The harbor is at (2, 5).

The old windmill is 4 units to the right and 3 units down from the harbor.

What are the coordinates of the old windmill?

A (6, 8)

B (6, 2)

C (2, 2)

D (5, 1)

E (1, 6)

5

What is the missing number in this sequence?

412 395 378 _____ 344

A 360

B 361

C 362

D 363

E 364

6

A large box has a volume of 192 cubic centimetres. It is being completely filled with identical smaller wooden cubes. Each smaller cube has a side length of 2 cm.

How many small cubes will completely fill the large box?

A 12

B 24

C 48

D 64

E 96

7

Anish was 1.54 metres tall.
He grew another 8 centimetres.

How tall was he then in metres?

- A** 1.548 m **B** 1.62 m **C** 2.34 m **D** 1.58 m **E** 1.602 m
-

8

A large regular grid contains 12 equal-sized squares. 4 squares are shaded black and 2 are shaded grey.

What fraction of the whole shape is unshaded?

- A** $\frac{1}{3}$ **B** $\frac{1}{4}$ **C** $\frac{1}{2}$ **D** $\frac{5}{12}$ **E** $\frac{7}{12}$
-

9

Write the correct number in the box.

$0.45 \times \blacksquare = 45$

- A** 0.1 **B** 10 **C** 100 **D** 1000 **E** 0.01

10

$$3a - 7 = 20$$

$$a = ?$$

A 9

B 7

C 6

D 11

E 8

11

A large bottle holds 2.5 litres of lemonade.

A server pours out four glasses of lemonade, each containing 450 ml.

How much lemonade is left in the bottle?

A 700 ml

B 0.5 l

C 1.8 l

D 1200 ml

E 0.75 l

12

A bar chart tracks how many hours Sarah spent reading each day. The vertical scale increases in increments of 2.5 hours. Her bar for Monday reaches up to 3 increments high.

How many hours did she spend reading on Monday?

A 5 hours

B 6 hours

C 7.5 hours

D 8 hours

E 10 hours

13

Tom saves £3.40 every week.

How many weeks will it take him to save exactly £51?

- A 12 weeks B 15 weeks C 17 weeks D 20 weeks E 22 weeks
-

14

A large rectangular floor measures 12 cm by 8 cm. Small decorative tiles measure 3 cm by 2 cm each.

How many small tiles will fit perfectly into the large rectangle?

- A 12 B 16 C 18 D 20 E 24
-

15

The perimeter of a square garden is 48 metres.

What is the total area of the garden?

- A 48 sq m B 96 sq m C 144 sq m D 12 sq m E 160 sq m

16

240 people voted for their favourite weekend hobby. On a pie chart, the sector for 'Swimming' has an angle of 60 degrees.

How many people voted for swimming?

A 30

B 40

C 50

D 60

E 80

17

A flight left at 21:45.

It arrived at 01:15 the next morning.

How long did the journey take, in minutes?

A 150 mins

B 190 mins

C 210 mins

D 230 mins

E 250 mins

18

Point X has coordinates $(-3, 4)$.

Point X is reflected across the y-axis.

What will be its new coordinates?

A $(3, 4)$

B $(-3, -4)$

C $(3, -4)$

D $(4, -3)$

E $(-4, 3)$

19

I took 2.5 hours to travel a total distance of 150 miles.

What was my average speed, in miles per hour?

A 50 mph

B 55 mph

C 60 mph

D 65 mph

E 70 mph

20

Which of the following statements is true for all parallelograms?

A All four sides equal.

B Four right angles.

C Diagonals cross at 90° .

D Opposite sides parallel & equal.

E One line of symmetry.

21

Bus 2 from High Street departs 09:02.

Bus 2 arrives at Park Lane at 09:29.

How many minutes does it take to get from High Street to Park Lane?

A 15 mins

B 27 mins

C 32 mins

D 44 mins

E 57 mins

22

What percentage of 80 kg is 1600 grams?

A 2%

B 5%

C 12%

D 20%

E 25%

23

A conversion graph shows that 5 miles equals 8 kilometres.

Using this, how many kilometres are equal to 35 miles?

A 40 km

B 48 km

C 56 km

D 64 km

E 72 km

24

A store reduces the price of a mountain bike by 15%.

The original price was £180.

How much money is saved with this discount?

A £15.00

B £18.00

C £24.00

D £27.00

E £32.00

25

Angle X is on a straight line alongside an angle measuring 63 degrees.

What is the size of angle X?

- A 27 degrees B 117 degrees C 127 degrees D 207 degrees E 297 degrees
-

26

Write the correct number in the box.

■ $\div 1.5 = 60$

- A 40 B 45 C 75 D 90 E 120
-

27

A student's spelling test scores out of 20:

Monday 14 Tuesday 19 Wednesday 15 Thursday 13 Friday 19

What was the mean average score?

- A 15 B 16 C 17 D 18 E 19

28

A gardener is paid £4.50 for every 10 square metres of lawn mowed.

How much are they paid for mowing 35 square metres?

A £13.50

B £15.75

C £16.00

D £18.25

E £22.50

29

$0.072 \times 1000 =$

A 0.72

B 7.2

C 72

D 720

E 7200

30

In 2010, a library loaned 15,000 books.

By 2020, the number had tripled.

How many books were loaned in 2020?

A 30,000

B 45,000

C 50,000

D 60,000

E 75,000

31

A crate has 84 items in it.
The inspector removes $\frac{4}{7}$ of them.

How many items did the inspector remove?

A 12

B 36

C 44

D 48

E 52

32

A commuter train travels 634 miles in one week.

How many total miles does it cover over 24 weeks?

A 14,216

B 15,216

C 15,224

D 15,316

E 16,216

33

8 27 64

The three numbers above share a clear mathematical property.

Select the statement that correctly describes how they are alike.

A All square numbers. **B** All prime numbers. **C** All cube numbers. **D** All divisible by 8. **E** All odd numbers.

34

The temperature in Chicago was 4°C in the morning.

By midnight, the temperature had dropped by 11°C .

What was the temperature at midnight?

A -6°C

B -7°C

C -8°C

D 15°C

E -11°C

35

Put these fractions in order of size, starting with the largest first.

$\frac{2}{3}$ $\frac{5}{6}$ $\frac{7}{12}$ $\frac{3}{4}$

A $\frac{5}{6}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{7}{12}$ **B** $\frac{7}{12}$, $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$ **C** $\frac{5}{6}$, $\frac{2}{3}$, $\frac{3}{4}$, $\frac{7}{12}$ **D** $\frac{3}{4}$, $\frac{5}{6}$, $\frac{2}{3}$, $\frac{7}{12}$ **E** $\frac{2}{3}$, $\frac{5}{6}$, $\frac{3}{4}$, $\frac{7}{12}$

36

A worker packs 8 identical delivery boxes.

Each box weighs 450 grams.

How many KILOGRAMS does this order weigh altogether?

A 3.2 kg

B 3.6 kg

C 36 kg

D 3600 kg

E 4.5 kg

37

What is $2^3 + 5^2$?

A 16

B 21

C 31

D 33

E 29

38

40 sweets are shared between Lucas and Mia in the ratio 3 : 2.

How many sweets does Lucas receive?

A 16 sweets

B 20 sweets

C 24 sweets

D 28 sweets

E 30 sweets

39

What is 35% of 80?

A 24

B 26

C 28

D 30

E 32

40

In a Venn Diagram, the left circle contains multiples of 6 and the right circle contains multiples of 8.

Which number must go into the overlapping shaded section?

A 12

B 16

C 18

D 24

E 32

41

$7.3 - 0.456 = ?$

A 6.844

B 6.854

C 7.156

D 7.244

E 6.944

42

On a number line from -100 to $+100$,

Which value lies exactly halfway between -60 and $+20$?

A -40

B -30

C -20

D -10

E 0

43

Given that: $48 \times 23 = 1104$

Select the correct value for: $11.04 \div 0.23 = \blacksquare$

A 0.48

B 4.8

C 48

D 480

E 4800

44

A school needs to transport 435 students.

Each shuttle bus holds a maximum of 32 passengers.

How many buses must be booked so that every student has a seat?

A 13 buses

B 14 buses

C 15 buses

D 16 buses

E 17 buses

45

An outdoor field day lasts 10 hours.

It rained for 30% of the duration and was cloudy for 50%.

For how many hours was it completely clear?

A 2 hours

B 3 hours

C 4 hours

D 5 hours

E 6 hours

46

Put the correct number in the box.

$$35 \times 98 = 3500 - \blacksquare$$

A 35

B 70

C 105

D 140

E 175

47

A class of 30 students voted for preferred lunch.

Block heights: Sandwich 2, Salad 1, Hot Wrap 2, Pizza 5.

How many children voted for Pizza?

A 5 children

B 10 children

C 12 children

D 15 children

E 18 children

48

The length of each side of a square layout is precisely tripled.

How many times larger does the total area become?

A 3 times

B 6 times

C 9 times

D 12 times

E 27 times

49

Chloe's mean score after 4 rounds was 42 points.
After the 5th round, her overall mean dropped to 40 points.

How many points did Chloe score in her fifth round?

- A** 32 points **B** 34 points **C** 36 points **D** 38 points **E** 40 points
-

50

The coordinates of 4 vertices of a quadrilateral are:
 $(a-4, b)$ (a, b) $(a-4, b+4)$ $(a, b+4)$

What specific geometric shape is formed?

- A** Rectangle **B** Kite **C** Trapezium **D** Parallelogram **E** Square